Lesson 03 material: Experimental and Clinical Psychologists

1.6: Experimental and Clinical Psychologists

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Learning Objectives

1. Define the clinical practice of psychology and distinguish it from experimental psychology.
2. Explain how science is relevant to clinical practice.
3. Define the concept of an empirically supported treatment and give some examples.

Who Conducts Scientific Research in Psychology?

Experimental Psychologists

Scientific research in psychology is generally conducted by people with doctoral degrees (usually the **doctor of philosophy [Ph.D.]**) and master's degrees in psychology and related fields, often supported by research assistants with bachelor's degrees or other relevant training. Some of them work for government agencies (e.g., doing research on the impact of public policies), national associations (e.g., the American Psychological Association), non-profit organizations (e.g., National Alliance on Mental Illness), or in the private sector (e.g., in product marketing and development; organizational behavior). However, the majority of them are college and university faculty, who often collaborate with their graduate and undergraduate students. Although some researchers are trained and licensed as clinicians for mental health work—especially those who conduct research in clinical psychology—the majority are not. Instead, they have expertise in one or more of the many other subfields of psychology: behavioral neuroscience, cognitive psychology, developmental psychology, personality psychology, social psychology, and so on. Doctoral-level researchers might be employed to conduct research full-time or, like many college and university faculty members, to conduct research in addition to teaching classes and serving their institution and community in other ways.

Of course, people also conduct research in psychology because they enjoy the intellectual and technical challenges involved and the satisfaction of contributing to scientific knowledge of human behavior. You might find that you enjoy the process too. If so, your college or university might offer opportunities to get involved in ongoing research as either a research assistant or a participant. Of course, you might find that you do not enjoy the process of conducting scientific research in psychology. But at least you will have a better understanding of where scientific knowledge in psychology comes from, an appreciation of its strengths and limitations, and an awareness of how it can be applied to solve practical problems in psychology and everyday life.

Scientific Psychology Blogs

A fun and easy way to follow current scientific research in psychology is to read any of the many excellent blogs devoted to summarizing and commenting on new findings. Among them are the following:

Research Digest, http://digest.bps.org.uk/

Talk Psych, http://www.talkpsych.com/

Brain Blogger, http://brainblogger.com/

Mind Hacks, http://mindhacks.com/

PsyBlog, http://www.spring.org.uk

You can also browse to http://www.researchblogging.org, select psychology as your topic, and read entries from a wide variety of blogs.

Clinical Psychologists

Psychology is the scientific study of behavior and mental processes. But it is also the application of scientific research to "help people, organizations, and communities function better" (American Psychological Association, 2011)[1]. By far the most common and widely known application is the **clinical practice of psychology**—the diagnosis and treatment of psychological disorders and related problems. Let us use the term *clinical practice* broadly to refer to the activities of clinical and counseling psychologists, school psychologists, marriage and family therapists, licensed clinical social workers, and others who work with people individually or in small groups to identify and help address their psychological problems. It is important to consider the relationship between scientific research and clinical practice because many students are especially interested in clinical practice, perhaps even as a career.

The main point is that psychological disorders and other behavioral problems are part of the natural world. This means that questions about their nature, causes, and consequences are empirically testable and therefore subject to scientific study. As with other questions about human behavior, we cannot rely on our intuition or common sense for detailed and accurate answers. Consider, for example, that dozens of popular books and thousands of websites claim that adult children of alcoholics have a distinct personality profile, including low self-esteem, feelings of powerlessness, and difficulties with intimacy. Although this sounds plausible, scientific research has demonstrated that adult children of alcoholics are no more likely to have these problems than anybody else (Lilienfeld et al., 2010)[2]. Similarly, questions about whether a particular psychotherapy is effective are empirically testable questions that can be answered by scientific research. If a new psychotherapy is an effective treatment for depression, then systematic observation should reveal that

depressed people who receive this psychotherapy improve more than a similar group of depressed people who do not receive this psychotherapy (or who receive some alternative treatment). Treatments that have been shown to work in this way are called **empirically supported treatments**.

Empirically Supported Treatments

An empirically supported treatment is one that has been studied scientifically and shown to result in greater improvement than no treatment, a placebo, or some alternative treatment. These include many forms of psychotherapy, which can be as effective as standard drug therapies. Among the forms of psychotherapy with strong empirical support are the following:

- **Acceptance and committment therapy (ACT)**. for depression, mixed anxiety disorders, psychosis, chronic pain, and obsessive-compulsive disorder.
- **Behavioral couples therapy.** For alcohol use disorders.
- **Cognitive behavioral therapy (CBT).** For many disorders including eating disorders, depression, anxiety disorders, etc.
- **Exposure therapy.** For post-traumatic stress disorder and phobias.
- **Exposure therapy with response prevention.** For obsessive-compulsive disorder.
- **Family-based treatment.** For eating disorders.

For a more complete list, see the following website, which is maintained by Division 12 of the American Psychological Association, the Society for Clinical Psychology: http://www.div12.org/psychological-treatments

Many in the clinical psychology community have argued that their field has not paid enough attention to scientific research—for example, by failing to use empirically supported treatments—and have suggested a variety of changes in the way clinicians are trained and treatments are evaluated and put into practice. Others believe that these claims are exaggerated and the suggested changes are unnecessary (Norcross, Beutler, & Levant, 2005)[3]. On both sides of the debate, however, there is agreement that a scientific approach to clinical psychology is essential if the goal is to diagnose and treat psychological problems based on detailed and accurate knowledge about those problems and the most effective treatments for them. So not only is it important for scientific research in clinical psychology to continue, but it is also important for clinicians who never conduct a scientific study themselves to be scientifically literate so that they can read and evaluate new research and make treatment decisions based on the best available evidence.

References

1. American Psychological Association. (2011). *About APA*. Retrieved from http://www.apa.org/about↵
2. Lilienfeld, S. O., Lynn, S. J., Ruscio, J., & Beyerstein, B. L. (2010). *50 great myths of popular psychology*. Malden, MA: Wiley-Blackwell. ↵
3. Norcross, J. C., Beutler, L. E., & Levant, R. F. (Eds.). (2005). *Evidence-based practices in mental health: Debate and dialogue on the fundamental questions*. Washington, DC: American Psychological Association. ↵

Links on the original page

- Web page (div12.org): http://www.div12.org/psychological-treatments
- Web page (apa.org): http://www.apa.org/about

1.7: The Science of Psychology (Summary)

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Key Takeaways

- Knowledge is acquired in many ways including intuition, authority, rationalism, empiricism, and the scientific method
- Science is a general way of understanding the natural world. Its three fundamental features are systematic empiricism, empirical questions, and public knowledge.
- Psychology is a science because it takes the scientific approach to understanding human behavior.
- Pseudoscience refers to beliefs and activities that are claimed to be scientific but lack one or more of the three features of science. It is important to distinguish the scientific approach to understanding human behavior from the many pseudoscientific approaches.
- Psychologists conduct research in order to describe basic phenomenon, to make predictions about future behaviors, and to explain the causes of behavior.
- Basic research is conducted to learn about human behavior for its own sake, and applied research is conducted to solve some practical problem. Both are valuable, and the distinction between the two is not always clear-cut.
- People's intuitions about human behavior, also known as folk psychology, often turn out to be wrong. This is one primary reason that psychology relies on science rather than common sense.
- Researchers in psychology cultivate certain critical-thinking attitudes. One is skepticism. They search for evidence and consider alternatives before accepting a claim about human behavior as true. Another is tolerance for uncertainty. They withhold judgment about whether a claim is true or not when there is insufficient evidence to decide.
- Scientific research in psychology is conducted mainly by people with doctoral degrees in psychology and related fields, most of whom are college and university faculty members. They do so for professional and for personal reasons, as well as to contribute to scientific knowledge about human behavior. Most psychologists are experimental psychologists and they conduct research.
- The clinical practice of psychology—the diagnosis and treatment of psychological problems—is one important application of the scientific discipline of psychology.
- Scientific research is relevant to clinical practice because it provides detailed and accurate knowledge about psychological problems and establishes whether treatments are effective.

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Exercises

- Practice: Consider three things you know and determine how you acquired that knowledge (authority, intuition, rationalism, empiricism, the scientific method).
- Practice: Try to generate different research questions to describe, predict, and explain a phenomenon that interests you.
- Practice: Based on your own experience or on things you have already learned about psychology, list three basic research questions and three applied research questions of interest to you.
- Practice: List three empirical questions about human behavior. List three nonempirical questions about human behavior.
- Practice: For each of the following intuitive beliefs about human behavior, list three reasons that it might be true and three reasons that it might not be true:
  - You cannot truly love another person unless you love yourself.
  - People who receive "crisis counseling" immediately after experiencing a traumatic event are better able to cope with that trauma in the long term.
  - Studying is most effective when it is always done in the same location.

Watch the following video, in which psychologist Scott Lilienfeld talks about confirmation bias, tunnel vision, and using evidence to evaluate the world around us:

Video: https://youtu.be/Eut8jMfSA_k

Dr. Scott Lilienfeld talks about confirmation bias, tunnel vision, and using evidence to evaluate the world around us: https://youtu.be/Eut8jMfSA_

- Discussion: Consider the following psychological claim. "People's choice of spouse is strongly influenced by their perception of their own parents. Some choose a spouse who is similar in some way to one of their parents. Others choose a spouse who is different from one of their parents." Is this claim falsifiable? Why or why not?
- Discussion: People sometimes suggest that psychology cannot be a science because either (a) human behavior cannot be predicted with perfect accuracy or (b) much of its subject matter (e.g., thoughts and feelings) cannot be observed directly. Do you agree or disagree with each of these ideas? Why?
- Watch the following video by PHD Comics for an overview of open access publishing and why it matters:

Video: https://youtu.be/L5rVH1KGBCY

Open Access Explained! https://youtu.be/L5rVH1KGBCY

- Discussion: Some clinicians argue that what they do is an "art form" based on intuition and personal experience and therefore cannot be evaluated scientifically. Write a paragraph about how satisfied you would be with such a clinician and why from each of three perspectives:
  - a potential client of the clinician
  - a judge who must decide whether to allow the clinician to testify as an expert witness in a child abuse case
  - an insurance company representative who must decide whether to reimburse the clinician for their services
- Practice: Create a short list of questions that a client could ask a clinician to determine whether they pay sufficient attention to scientific research.

Links on the original page

- Web page (apa.org): http://www.apa.org/about
- Video (YouTube): https://youtu.be/Eut8jMfSA_
- Video (YouTube): https://youtu.be/L5rVH1KGBCY